

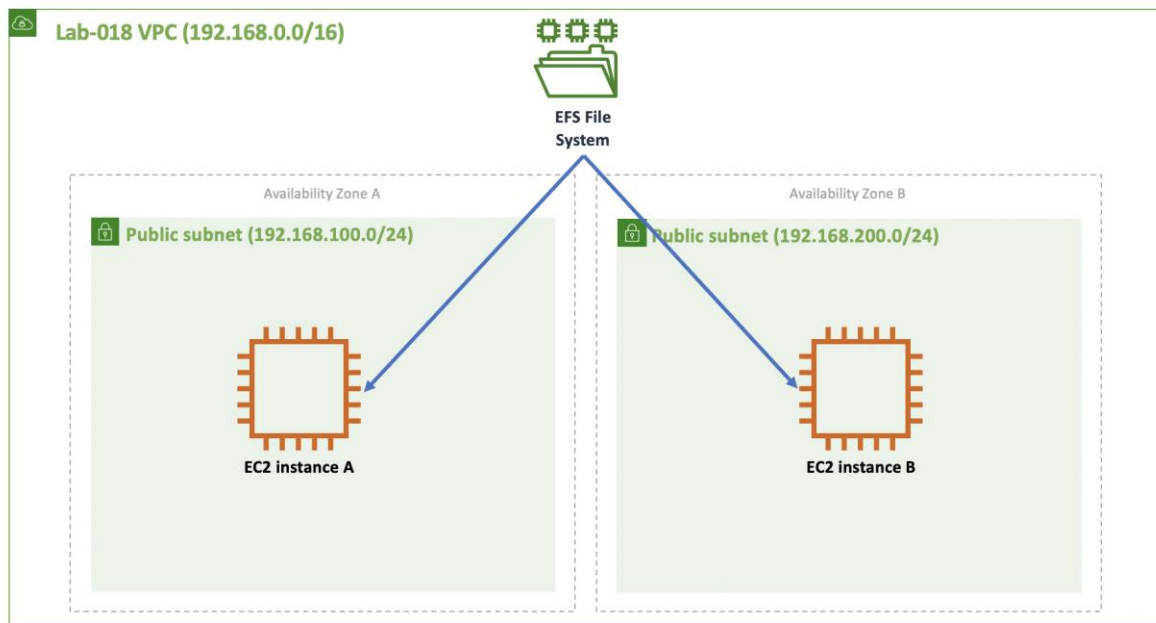
Lab-018

Creating and Sharing an NFS File System via EFS

Goal

The goal of this lab is to illustrate how to create and share an NFS file system using AWS [EFS](#) service.

Architecture Diagram



Overview

Create two EC2 instances in different AZs (you can use public subnets). Create an NFS file system using EFS and mount it using one of the EC2 instances. Create some files for testing purposes. Then try to mount and access the file system from the other EC2 instance. Note that the access should work simultaneously.

Step 1 - VPC Setup

Make sure your VPC is configured to enable mounting using DNS names. You can do that by going to your VPC's action menu and selecting *Edit DNS Resolution* and *Edit DNS Hostnames*. Make sure both are set to Yes.

Step 2 - Create Security Group

Create a security group named *nfs-access* to allow access to the NFS file system to members of the security group.

First Create the security group.

EC2 > Security Groups > Create security group

Create security group [Info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [Info](#)

nfs-access

Name cannot be edited after creation.

Description [Info](#)

nfs-access

VPC [Info](#)

vpc-080c2ae1ce3f75272 (myVPC)

Then edit its inbound rule to allow access from members of the group.

EC2 > Security Groups > sg-056a90da9efc23b63 - nfs-access > Edit inbound rules

Edit inbound rules [Info](#)

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules [Info](#)

Type Info	Protocol Info	Port range Info	Source Info	Description - optional Info	
NFS	TCP	2049	Custom Q sg-056a90da9efc23b63		Delete

Add rule

NOTE: Any edits made on existing rules will result in the edited rule being deleted and a new rule created with the new details. This will cause traffic that depends on that rule to be dropped for a very brief period of time until the new rule can be created.

Cancel Preview changes Save rules

Step 3 - Launch EC2 Instances

Launch two EC2 instances, each on its own AZ. Use the user-data.sh to install Amazon's EFS utility package. Make sure your instances are members of the *nfs-access* security group and are also accessed via ssh.

Step 4 - Create an NFS File System

Go to *Storage - EFS* and click *Create file system*.

Configure network access

An Amazon EFS file system is accessed by EC2 instances running inside one of your VPCs. Instances connect to a file system by using a network interface called a mount target. Each mount target has an IP address, which we assign automatically or you can specify.

VPC vpc-080c2ae1ce3f7527... ⓘ

Create mount targets

Instances connect to a file system by using mount targets you create. We recommend creating a mount target in each of your VPC's Availability Zones so that EC2 instances across your VPC can access the file system.

	Availability Zone	Subnet	IP address	Security groups
☒	us-west-1a	subnet-08b070c3063ae5afa - publicA	Automatic	sg-056a90da9efc23b63 - nfs-access
☒	us-west-1c	subnet-085ed04125ea8ae2b - publicB	Automatic	sg-056a90da9efc23b63 - nfs-access

Cancel Next Step

Copy the EFS file system ID.

File systems

Create file system

Actions ▾

		Name	File system ID	Metered size
☐	▶		fs-02faaa1b	18.0 KiB
☒	▶	myEFS	fs-9ee0b987	6.0 KiB

Step 5 - Mount the NFS File System

Access one of the EC2 instances using ssh. Create a folder to be the mounting point (let's say *data*). Then using your EFS file system ID (mine was *fs-9ee0b987*) issue the command:

```
sudo mount -t efs fs-9ee0b987:/ data
```

You should be able to access the file system. Create a few testing files.

Test and Validation

Connect to the second EC2 instance (the one running on the other AZ), mount the EFS file system and see if you can access the files created earlier. Make sure you test simultaneous access (both EC2 accessing the file system).